CASE STUDY – 02

"Temperature Monitoring System"

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Problem Statement:

- Build a Python code to display messages according to the temperature received from an assumed IoT system.
- Accept max and min limit temperature.
- Generate random values for temperature at every 2 second interval.
- Compare with the limits to display appropriate value.

PYTHON CODE

```
import random   # to generate random temperature
import time     # to add 2 second delay
# take input of temperature limits from user
min_limit = float(input("Enter minimum temperature limit: "))
max_limit = float(input("Enter maximum temperature limit: "))
# Run continuously
while True:
   # Generate random temperature between 0 and 100
   temperature = random.randint(0, 100)
   print("Current Temperature:", temperature)
   # Compare temperature with limits
   if temperature > max_limit:
      print("Alert: Temperature is too high\n")
   elif temperature < min_limit:
      print("Alert: Temperature is too low\n")
   else:
      print("Temperature is within acceptable limit\n")

   time.sleep(2)  # Wait for 2 seconds
```